

Massive Hemoptysis in COPD Exacerbation

Section 1: Case Summary

Scenario Title:	COPDe with Massive Hemoptysis
Keywords:	Hemoptysis, Bronchiectasis, Intubation, Anticoagulation, COPD Exacerbation
Brief Description of Case:	This is a case of a 57-year-old male with former smoking history, COPD on home O2, atrial fibrillation on anticoagulation coming into a community emergency department (ED) with symptoms consistent with COPD exacerbation. His initial presentation will respond to traditional therapies for COPD exacerbation and positive pressure ventilation. During his ED stay, however, the patient will develop a coughing fit and hemoptysis with airway compromise, requiring learners to secure the airway, reverse anticoagulation, and manage massive hemoptysis.

Goals and Objectives	
Educational Goal:	Recognize massive hemoptysis as an airway emergency with effective airway securement plan and adjunct management
Objectives: (Medical and CRM)	1. Recognize decompensation of patient and need for definitive airway 2. Convey priorities and plan of airway securement via shared mental model with team members 3. Recognize life threatening bleeding and reverse anticoagulation 4. Consultation of appropriate specialists for continued management of massive hemoptysis

Learners, Setting and Personnel			
Target Learners:	☐ Junior Learners	☒ Senior Learners	☒ Staff
	☒ Physicians	☒ Nurses	☒ RTs
	☒ Other Learners: Anesthesia		
Location:	☒ Sim Lab	☒ In Situ	☐ Other:
Recommended Number of Facilitators:	Instructors: 1		
	Sim Actors: 2 (Nurse and RT)		
	Sim Techs: 1		

Scenario Development	
Date of Development:	2/2/2022
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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Jackson Boggard		Age: 57	Gender: Male	Weight: 100 kg	
Presenting complaint: Cough and shortness of breath					
Temp: 38.1 C	HR: 124 bpm	BP: 162/94	RR: 22	O ₂ Sat: 90%	FiO ₂ : 3L NC
Cap glucose (if asked): 7.3			GCS: 15		
Triage note: Per EMS Report: Patient called EMS c/o SOB and cough, progressively worsening over the last three days. The patient is coming from home, where he was found sitting in his couch in mild respiratory distress. The patient is brought to your resuscitation room.					
Allergies: NKDA					
Past Medical History: COPD (3 L NC at home) HTN HLD Atrial Fibrillation Hx of smoking 2 pack per day x 25 years Received COVID vaccination and booster			Current Medications: Fluticasone with Salmeterol 115 mcg/21 mcg HFA Albuterol HFA Amlodipine 5 mg Atorvastatin 40 mg Warfarin 5 mg		

Section 2B: Extra Patient Information

A. Further History	
<i>Include any relevant history not included in triage note above. What information will only be given to learners if they ask? Who will provide this information (mannequin's voice, sim actors, SP, etc.)?</i>	
1) Patient is normally on 3 L NC at home 2) Cough has been more productive of green sputum over the last 2 days 3) Hospitalized for COPD exacerbation three times in the last year, but no history of intubation or ICU stay. 4) INR check three days prior was 2.3 5) Patient only feels short of breath when coughing fits or moving around the home 6) No history of PE/DVT, no recent travel, no sick contacts	
B. Physical Exam	
<i>List any pertinent positive and negative findings</i>	
Cardio: -CP	Neuro: - HA, dizziness
Resp: +SOB, +Cough, +Wheezing	Head & Neck: +Rhinorrhea , - Sore Throat
Abdo: -Abdominal Pain, - n/v/d	MSK/skin: -Rashes
Other: No leg swelling, paroxysmal nocturnal dyspnea, orthopnea	

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Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin <i>Adult</i>
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Standard supplemental oxygen supplies CPAP or BiPAP Ventilator Intubation supplies Cricothyrotomy Kit Suction Canister, suction tubing, and Yankauer Ultrasound
C. Required Medications
Albuterol/Ipratropium Steroid (i.e. methylprednisolone, prednisone, hydrocortisone) Antibiotics – Levofloxacin, Ceftriaxone, Cefotaxime Magnesium Vitamin K 4 Factor Prothrombin Complex Concentrate (PCC) Fresh Frozen Plasma (FFP) Tranexamic acid (TXA) Isotonic fluids (i.e. Normal Saline, Lactated Ringers, Plasmalyte) Packed Red Blood Cells RSI medications
D. Moulage
Diaphoretic (spray manikin)
E. Monitors at Case Onset
☒ Patient on monitor with vitals displayed ☐ Patient not yet on monitor
F. Patient Reactions and Exam
<i>Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format.</i> A) Coughing, but speaking full sentences on initial presentation B) Wheezing in all lung fields C) Rapid and irregular, bounding pulse D) GCS 15, moving 4 x extremities

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Section 4: Sim Actor and Standardized Patients

Sim Actor and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
Bedside ED Nurse	Assist with starting IVs, drawing labs, giving medications, updating changes in vitals. Skilled and helpful.
Respiratory Therapy	Assist with ABG, intubation, and ventilator set-up. Skilled and helpful.

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Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: Atrial Fibrillation HR: 124 bpm BP: 164/92 RR: 24 O ₂ SAT: 90% (3L NC) T: 38.1 °C GCS: 15	<i>Coughing, speaking full sentences in mild respiratory distress with tachypnea and audible wheezing.</i>	<u>Expected Learner Actions</u> ☐ Establish IVs x 2 ☐ Consider PPV/HFNC ☐ Draw labs, blood cultures ☐ EKG ☐ Salbutamol/ipratropium ☐ Corticosteroid ☐ Antibiotics ☐ CXR ☐ IV fluids	<u>Modifiers</u> - Rate control -> BP 80/55 - IVF -> HR 110 <u>Triggers</u> - COPD treatment given → 2. Improved COPD Exacerbation	
2. Improved COPD Exacerbation Rhythm: AF HR: 110 bpm BP: 142/84 RR: 20 O ₂ SAT: 94% (3L NC)	<i>Improved cough</i>	<u>Expected Learner Actions</u> ☐ Complete initial workup ☐ Interpret labwork appropriately	<u>Triggers</u> - After labwork interpretation or 5 min mark → 3. Start Hemoptysis	- ABG with electrolytes can be given at the start this stage
3. Start Hemoptysis Rhythm: Atrial Fibrillation HR: 140 bpm BP: 174/92 RR: 30 O ₂ SAT: 80% (3L NC) GCS: 14	<i>Coughing blood, 1 word dyspnea, Gurgling resps</i>	<u>Expected Learner Actions</u> ☐ Recognize hemoptysis as cause of deterioration ☐ CXR to localize bleed ☐ Position patient (upright to pre-oxygenate) ☐ Prepare double suction ☐ Prepare double set-up (cric) ☐ Perform RSI with R mainstem intubation	<u>Modifiers</u> - If no positioning after standard tracheal intubation, then O ₂ sats continue to drop <u>Triggers</u> - R mainstem intubation or standard tracheal intubation with left lat decubitus positioning → 4. Stable Massive Hemoptysis	- If learner reexamines lung exam, decreased breath sounds in left - Suction canister with 60 ccs blood upon initial coughing - If chest tube placed, no return of blood



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4. Stable Massive Hemoptysis Rhythm: Atrial Fibrillation HR: 110 bpm BP: 132/78 RR: Vent rate O2SAT: 100% FiO2 GCS: 2T	<i>Intubated</i>	<u>Expected Learner Actions</u> ☐ Reverse warfarin (PCC, Vitamin K) ☐ Type and Cross patient ☐ Confirm ETT location ☐ Foley, NGT after intubation ☐ Post-Intubation Sedation ☐ Consult ICU ☐ Order CTA Chest ☐ Consult interventional radiology, cardiothoracic surgery	<u>Modifiers</u> -If no reversal of warfarin, suction canister continues to increase of blood and becomes hypotensive -If ICU called prior to IR and/or CT surgery, intensivist asks, "Is the patient still bleeding? Is there someone else we should be calling?" <u>Triggers</u> -Consult ICU and other consultants → End Case	-Total blood in suction canister: 600 ccs
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Appendix A: Laboratory Results

<u>ABG</u> pH: 7.28 pCO ₂ : 75 mmHg pO ₂ : 88 mmHg HCO ₃ : 16 mmol/L Lactate: 3.0 mmol/L	
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Appendix B: ECGs, X-rays, Ultrasounds and Pictures

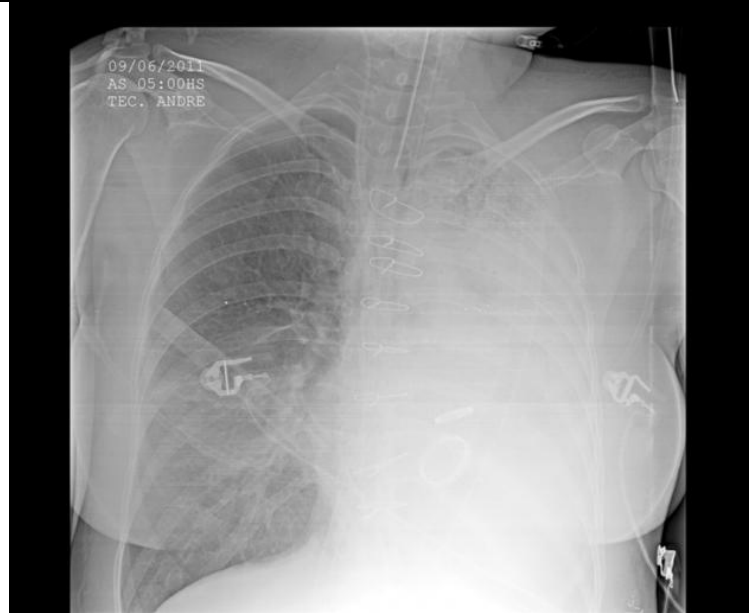


Initial CXR: Increased bronchovascular markings, hyperinflation, flattened diaphragms, and small heart
From Radiopaedia: (<https://radiopaedia.org/articles/chronic-obstructive-pulmonary-disease-1>)

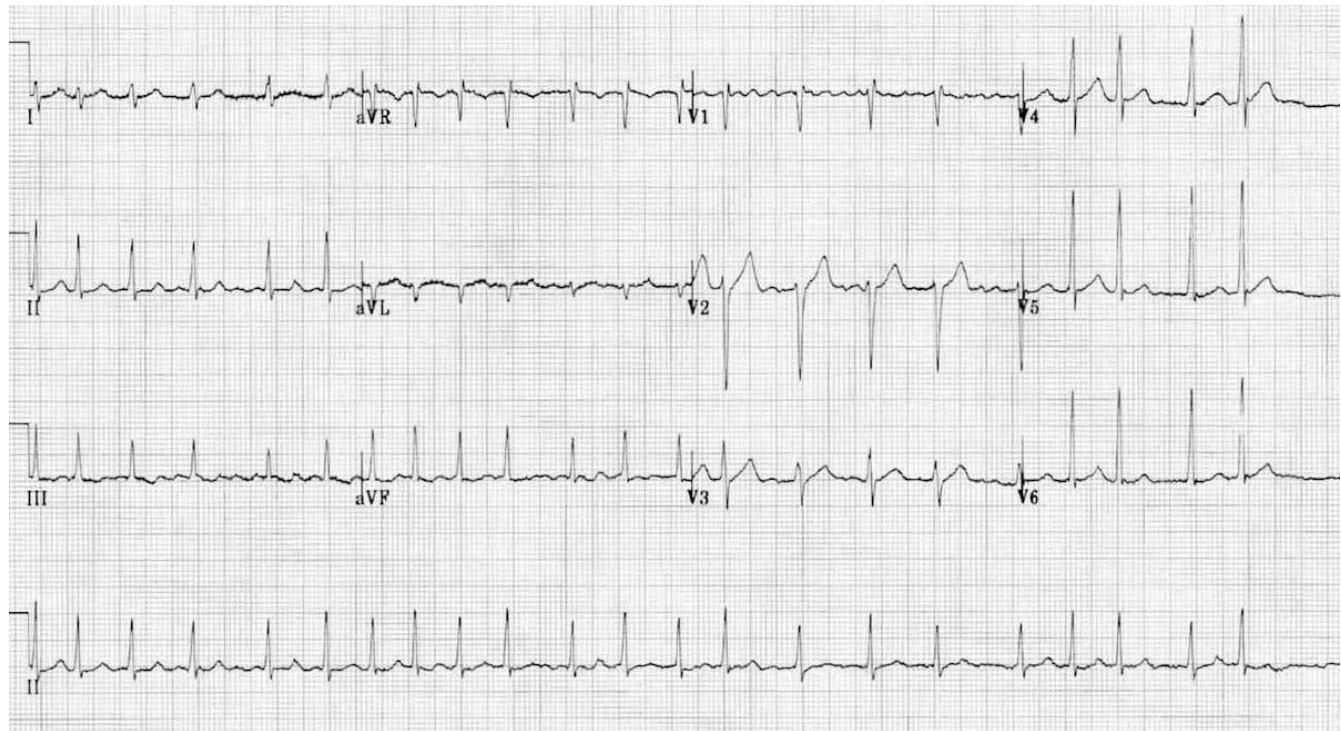


Massive Hemoptysis CXR: Left sided consolidation with increased bronchovascular markings
From FoamCAST: (<https://foamcast.org/tag/hemoptysis>)

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Post-Intubation CXR: Complete left hemithorax with endotracheal tube
From EMDaily: (<http://emdaily.cooperhealth.org/content/board-review-critical-care-0>)



Initial ECG: Atrial Fibrillation with rate ~135 bpm
From Life in the Fast Lane: (<https://litfl.com/wp-content/uploads/2018/08/ECG-Atrial-Fibrillation-4.jpg>)

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Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

Key Errors:

- 1 – Delayed recognition of massive hemoptysis and decompensation
- 2 – Not reversing bleeding diathesis
- 3 – Inappropriate positioning or no selective intubation
- 4 – Delayed consultation to consultants

Debriefing Topics:

- 1 – Definition of massive hemoptysis
- 2 – Etiologies of massive hemoptysis
- 3 – How to distinguish hemoptysis from hematemesis
- 4 – Airway and securement plan for massive hemoptysis
- 5 – Appropriate disposition in massive hemoptysis
- 6 – Review institutional emergency airway management plans
- 7 – Reversal of warfarin

References

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